

Roll No.....

END SEMESTER EXAMINATION, DEC 2025

Name of the Program: B.Tech(EE)
Name of the Course: Analog Electronics

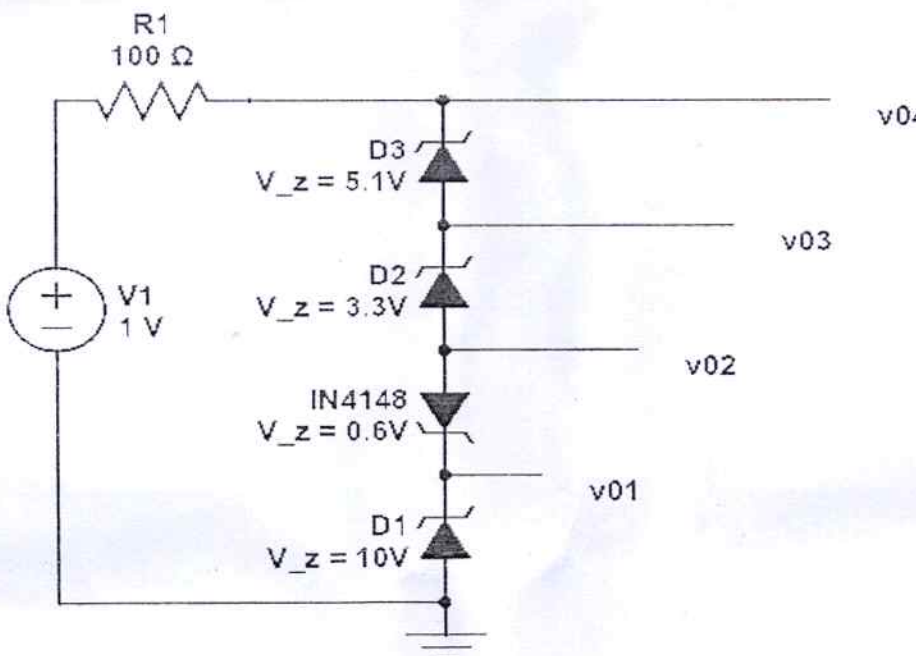
Semester: III
Course Code: TEE 304

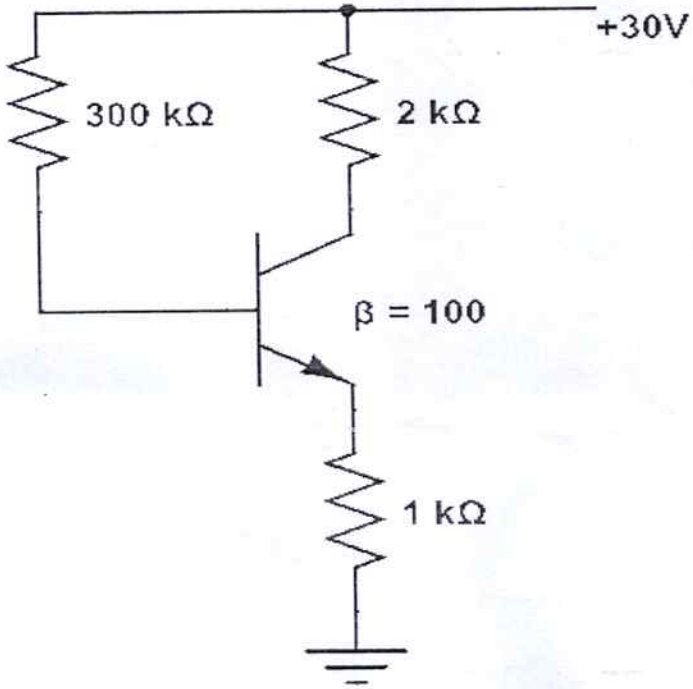
Time: 3 Hours

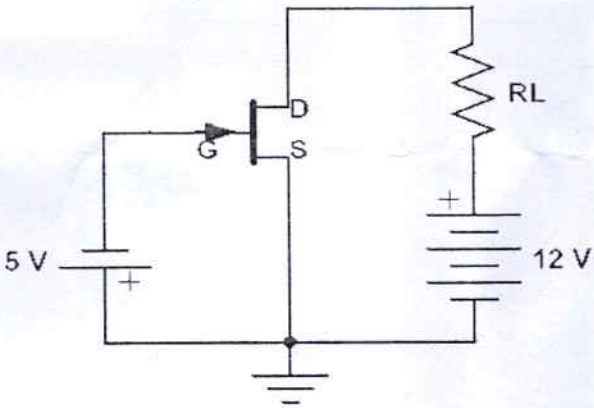
Maximum Marks: 100

Note:

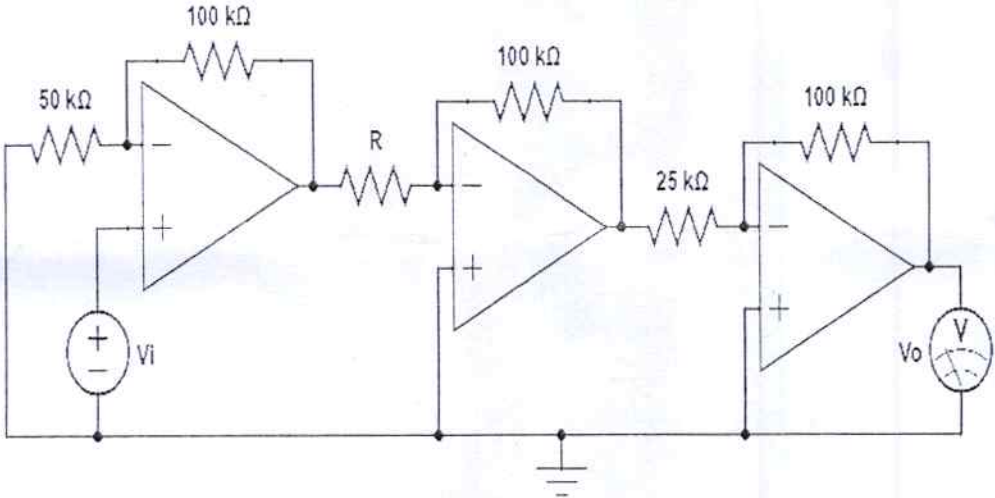
- (i) Answer questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each sub-question carries 10 marks

Q1	(20 marks)	
(a)	Draw the characteristics of the following: (i) Practical p-n junction diode (ii) An ideal diode Briefly explain the curves.	CO1
(b)	Write working of a bridge rectifier circuit. Draw wave shapes.	CO2
(c)	In the following circuit "Zeners connected in series", evaluate V_{01} , V_{02} , V_{03} and V_{04} 	CO2

Q2	(20 marks)	
(a)	A 5.0V stabilized power supply is required to be produced from a 12V DC power supply input source. The maximum power rating P_Z of the zener diode is 2.0W. Using the basic circuit of the zener regulator, calculate (i) The maximum current through the zener diode (ii) The minimum value of the series resistor R_s (iii) The load current I_L, if a load resistor of $1k\Omega$ is connected across the zener diode. (iv) The zener current at full load	CO3
(b)	For the circuit shown below, find (i) $I_{c(sat)}$, (ii) I_c, (iii) V_c, (iv) V_E and (v) V_{CE} 	CO3
(c)	Draw the circuit diagram of an NPN transistor employing Voltage Divider Bias. Discuss in brief this arrangement of biasing a BJT.	CO2

Q3	(20 marks)	
(a)	For the N-channel JFET shown in the circuit below has $V_{po} = 8V$ and $I_{DSS} = 12mA$. What would be the value of (i) V_{DS} at which pinch-off begins and (ii) value of I_D when V_{DS} is above pinch off and below the breakdown voltage?	CO4
		
(b)	An N-channel E-MOSFET has the following parameters ($I_{D(ON)} = 4mA$ at $V_{GS} = 10V$), and $V_{GS(th)} = 5V$. Calculate its drain current I_D at $V_{GS} = 8V$.	CO3
(c)	Draw the static characteristics ((i) Drain and (ii) transfer) of a DE MOSFET. Label these fully.	CO2

Q4	(20 marks)	
(a)	Fill in the blanks: (i) r_d is the ac resistance of the _____ between the drain (D) and the _____ terminals when FET is operating in the _____ region. (ii) r_d is given by the _____ of the drain characteristic in the _____ region. (iii) Since r_d is usually the output resistance of an FET, it may also be expressed as an output _____ Y_{os}. (iv) Trans-conductance g_m is the slope of the _____ characteristic of an FET. (v) Mathematically g_m is expressed as follows: $g_m =$ _____	CO3
(b)	Write the properties of an Ideal Operational Amplifier.	CO1
(c)	Draw the internal block diagram of an Operational Amplifier. Label all the internal building blocks.	CO1

Q5	(20 marks)	
(a)	In the cascade of ideal OpAmps, as shown below, if $V_i = 2V$ and $V_o = 30V$, determine the value of R. 	CO3
(b)	Draw the basic circuit of a square wave generator. Describe its working.	CO2
(c)	Enumerate various types of multi-vibrator circuits. Draw their basic circuit diagrams.	CO2